SECURITY INFRASTRUCTURE & THREAT DETECTION

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SECURITY ARCHITECTURE & INTRUSION DETECTION

7-LAYER DEFENSE ARCHITECTURE

L7 — PHYSICAL VALIDATION

`I2C Sensor Cross-Checking (Pi #6 — 10.30.0.25)`

Validates network sensor data against physical I2C sensor • Detection threshold: ± 1.0 PSI mismatch

L6 — DATA ENCRYPTION

`TLS/SSL • PostgreSQL SSL mode • Keycloak HTTPS • HAPI FHIR OAuth2 proxy`

All database connections require SSL/TLS • End-to-end encryption across all service interfaces

L5 — SECRETS MANAGEMENT

`HashiCorp Vault (10.30.0.58 : 8200)`

Centralized: DB credentials, API keys, encryption keys • 3-of-5 unseal keys (Shamir) • OIDC via Keycloak • Full audit log

L4 — IDENTITY & ACCESS CONTROL

`Keycloak SSO (10.30.0.57 : 8080) — Realm: hospital-homelab`

Single Sign-On for all services • OIDC: Grafana, Node-RED, Wazuh, Vault • Users: admin_user, doctor1

L3 — SIEM — SECURITY INFO & EVENT MGMT

`Wazuh SIEM (10.30.0.55 : 443) — 14 agents across all systems`

Event correlation • Anomaly detection • Real-time alerting • Forensic logging • Keycloak SSO

L2 — INTRUSION DETECTION SYSTEM (IDS)

`Suricata IDS (192.168.50.191 VLAN2 — bare metal laptop)`

Port mirroring: Port 15 mirrored to Port 7 • Signature-based + anomaly detection • EVE JSON → Filebeat → Elasticsearch 10.30.0.60:9200

L1 — NETWORK SEGMENTATION

`UDM-SE Gateway • VLAN30 (10.30.0.0/24) • VLAN2 (192.168.50.0/24) • Firewall`

Isolated attack lab (VLAN2) • Secure infrastructure (VLAN30) • VLAN2→VLAN30 BLOCKED (3 exceptions only)

Defense in Depth: Multiple overlapping controls ensure security even if one layer is bypassed • Each layer provides independent protection

INTRUSION DETECTION PIPELINE

NETWORK TRAFFIC — All Systems

- 9 VMs + 5 Cardiac Pis (Pi #1–5) + 2 Demo Pis (Pi #6–7)
- All traffic originates from VLAN30 & VLAN2

Physical Switch Port 15 • Pi #7 traffic

PORT MIRRORING — Port 15 mirrored to Port 7

- Complete copy of Pi #7 (192.168.50.100) traffic
- Passive tap — no impact on live network traffic

Mirrored copy to Suricata (promiscuous)

SURICATA IDS — 192.168.50.191 (VLAN2, bare metal)

- Monitoring interface: no IP (promiscuous mode, passive)
- Detection — Signature-based (Emerging Threats rules)
- Detection — Anomaly detection • Protocol analysis
- Alerts: HTTP sensor spoofing • Port scanning
- Alerts: Unauthorized access • Protocol violations
- Output: EVE JSON — timestamp, src/dst IP, severity, sig ID

Filebeat agent ships EVE JSON • 5-sec batch

ELASTICSEARCH — 10.30.0.60 : 9200

- Receives EVE JSON via Filebeat agent • TLS enabled
- Indexes: filebeat-* • Retention: unlimited (manual cleanup)
- Firewall exception: 192.168.50.191 → 10.30.0.60:9200

Grafana queries Elasticsearch index

GRAFANA IDS DASHBOARD — 10.30.0.53 : 3000

- Events over time (line graph) • Recent alerts (log panel)
- Top source/dest IPs (table) • Event types (bar chart)
- Network protocols (bar chart) • Total events counter
- Auto-refresh: 5 sec • Default: 1 hr • Keycloak SSO

IDS latency: < 1 second | Filebeat batch: 5 sec | Dashboard refresh: 5 sec | Retention: unlimited

SECURITY MONITORING & ACCESS CONTROL

WAZUH SIEM — 14-Agent Monitoring

`10.30.0.55 : 443 | Keycloak SSO | Event Correlation`

AGENTS (14 total)

Virtual Machines (9)

```
PostgreSQL   .50
MQTT Broker  .51
Node-RED     .52
Grafana      .53
Mirth Connect .54
Wazuh Mgr    .55
HAPI FHIR    .56
Keycloak     .57
Vault        .58
```

Raspberry Pi (5)

`Pi #1–5 (10.30.0.20–.24) — Cardiac monitors`

DETECTION CAPABILITIES

- ✓ Failed auth • Privilege escalation
- ✓ File integrity • Rootkit detection
- ✓ Anomalous connections • Service failures
- ✓ Config changes • Brute force attacks

RESPONSE ACTIONS

- Real-time alerts to dashboard
- Automatic incident ticket creation
- Forensic data collection
- Firewall integration for active response

KEYCLOAK SSO — Identity Provider

`10.30.0.57 : 8080 | Realm: hospital-homelab`

INTEGRATED SERVICES

- ✓ Grafana (dashboards)
- ✓ Node-RED (flow editor)
- ✓ Wazuh SIEM (security console)
- ✓ HashiCorp Vault (OIDC auth)

USER ROLES

- admin_user — Full system access
- doctor1 — Clinical dashboards

AUTHENTICATION FLOW

1. User accesses service
2. Redirect to Keycloak login
3. OIDC token issued
4. Service validates token
5. Session established

SECURITY FEATURES

- MFA ready
- Session timeout: 30 minutes
- Password policy enforcement
- Audit logging of all logins

HASHICORP VAULT — Secrets Management

`10.30.0.58 : 8200 | OIDC auth via Keycloak`

STORED SECRETS

- PostgreSQL credentials
- MQTT broker passwords
- API keys (HAPI FHIR)
- TLS certificates
- Encryption keys

SECURITY MODEL

- 5 unseal key shares (Shamir)
- 3-of-5 threshold required
- Auto-seals on VM restart
- Full audit logging

UNSEAL PROCESS (after restart)

```
vault operator unseal <key1>
vault operator unseal <key2>
vault operator unseal <key3>
# → Status: Unsealed
```

INTEGRATIONS

- Node-RED — DB credentials
- Mirth Connect — API keys
- Grafana — Datasource passwords
- Dynamic secret rotation

SECURITY MONITORING STATISTICS

14	7	100%	0%	< 1 sec	5 sec	∞
agents	layers	attacks	attacks	IDS	refresh	retention
Wazuh SIEM	Defense in Depth	detected	missed	latency	IDS dashboard	Elasticsearch